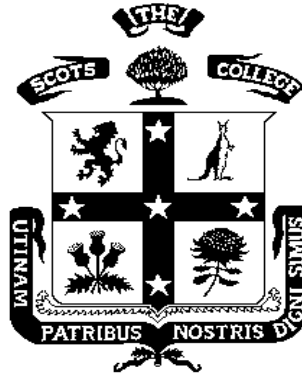


THE SCOTS COLLEGE Sydney



Extension One Mathematics HSC Task 3

Weighting 20%

7th June 2011

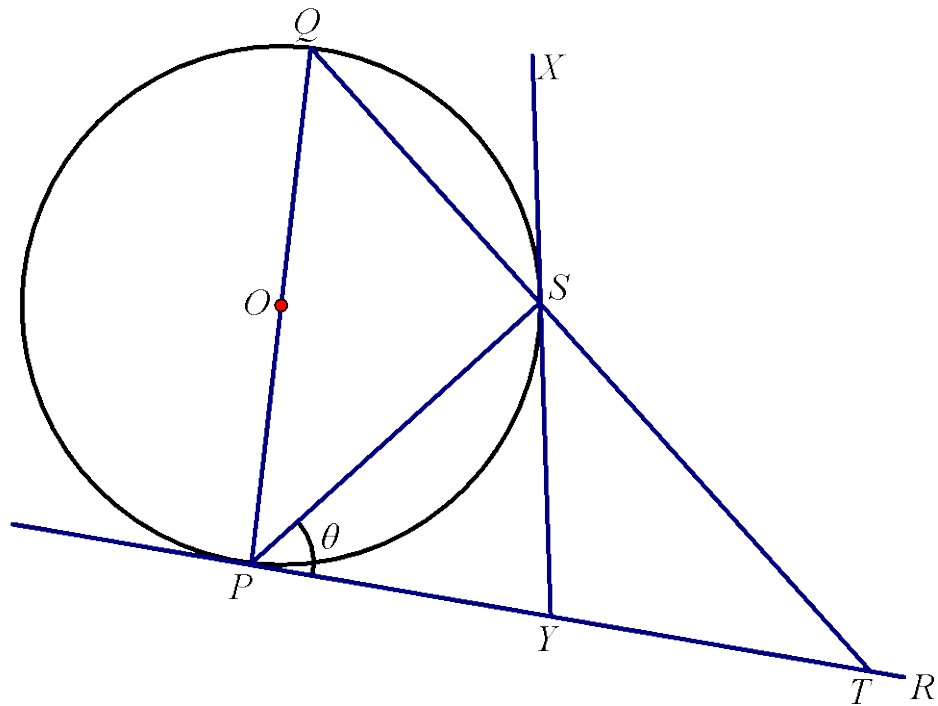
Total marks – 33

- Attempt all questions.

General Instructions

- **Working time – 45 Minutes.**
- Write using black or blue pen.
- Start a new page for every question
- Board-approved calculators may be used.
- All necessary working should be shown in every question.

QUESTION 1 (8 Marks)



PQ is the diameter of a circle with centre O . PR is the tangent to the circle at P . The line QST intersects the circle at S and PR at T . The tangent to the circle at S cuts PR at Y . Let

- Prove** that (you are not allowed to use $PY = YS$). [3]
- Prove** that $PY = YS$. [2]
- Prove** that Y is the mid point of PT . [3]

QUESTION 2**(4 Marks)****START A NEW PAGE**

Prove the following by mathematical induction

is divisible by 9 for all $n \geq 1$.

QUESTION 3**(5 Marks)****START A NEW PAGE**

A vessel is in the shape of an inverted, right cone, with an apex angle of 60° . Sand is being poured into the conical vessel at a constant rate of $48 \text{ cm}^3/\text{min}$.

- a) Show that the radius of the sand in the cone at any time is — , where h is the height of the sand in the vessel. [2]
- b) Find the rate at which the height of the sand in the vessel is increasing when the height is 12 cm. [3]

QUESTION 4**(8 Marks)****START A NEW PAGE**

An ice cube tray is filled with water at a temperature of 15°C . It is then placed in a freezer which has a constant temperature of -10°C . The water in the tray cools in the freezer at a rate that is proportional to the difference between the temperature of water, $W^\circ\text{C}$, at any time, and the temperature of the freezer, -10°C , so that W satisfies the equation

— , where k is the constant of proportionality.

- a) Show that ——— . [2]
- b) Hence show that ——— . [2]
- c) After 5 minutes, the temperature of the water in the tray is 5°C .
 - i. Find the value of k (to one decimal place) [2]
 - ii. Find the time that it takes for the water to cool down to -5°C (to the nearest minute). [2]

Question 5 continued on next page

QUESTION 5**(8 Marks)****START A NEW PAGE**

Three mathematicians decide to invest \$5000 into *PAL* Trust Fund in order to award an *annual* scholarship of \$500 to a deserving Mathematics student. The investment earns an interest at the rate of 9% *p.a.*, compounded *monthly*. The scholarship amount of \$500 is withdrawn at the end of each year after the monthly interest has been deposited.

- a) Find the amount remaining in the investment at the end of the first year, after awarding the scholarship. [2]
- b) Let A_n be the balance in the investment at the end of the n^{th} year, after the withdrawal of the scholarship amount for the year. If $R = 1.0075$, show that

_____ [3]

- c) Hence find the number of years for which the annual scholarship can be awarded (no scholarship is awarded if the balance in the investment falls below \$500). [3]

END OF PAPER

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

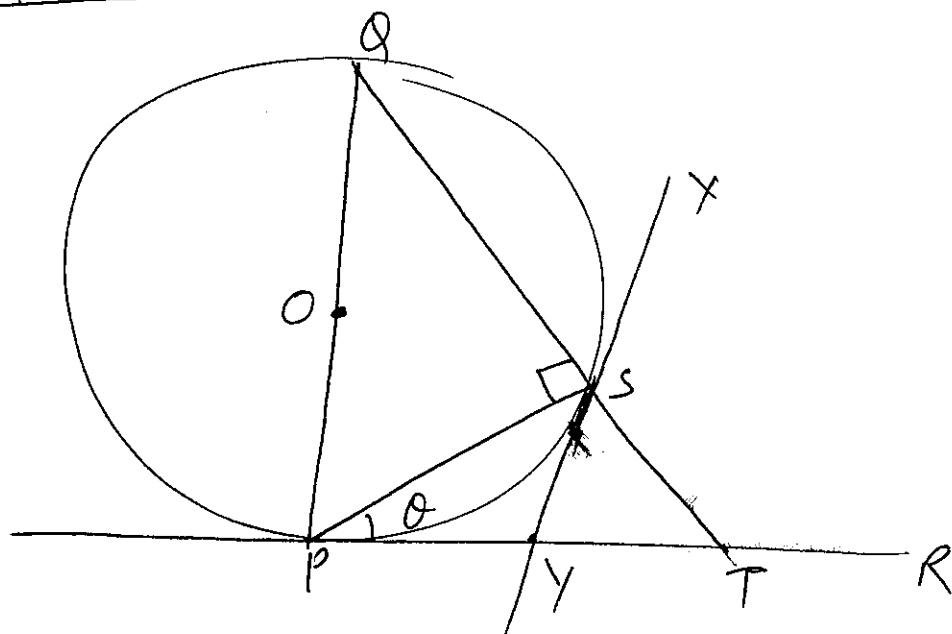
$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1

- (i) $\angle QPY = 90^\circ$ (line from centre to tangent is $\perp$ at the point of contact)
 $\therefore \angle QPS = 90 - \theta$
 $\angle XSQ = \angle QPS$ (alternate segment theorem)
 $= 90 - \theta$

$$\angle YST = \angle XSQ \text{ (vertically opposite angle)}$$

$$\therefore \angle YST = 90 - \theta.$$

- (ii) $\angle QSP = 90^\circ$ (angle in a semi-circle is a rt $\angle$)

$$\therefore \angle PST = 90^\circ \text{ (QST is a st. line)}$$

$$\therefore \angle PSY = \theta \text{ (}\because \angle YST = 90 - \theta\text{)}$$

$$= \angle SPY$$

$$\therefore \underline{PY = PS} \text{ (equal sides opposite to equal angles of a } \Delta \text{)}$$

- (iii) $\angle STP = 90 - \theta$ (angle sum of ΔPST)

$$\therefore \angle YST = \angle STY$$

$$\therefore SY = YT \text{ (equal sides opposite to equal angles of a } \Delta \text{)}$$

$$\therefore PY = YS = YT$$

$$\therefore \underline{Y \text{ is the mid pt of } PT.}$$

Q2:

$10^n + 3(4^{n+2}) + 5$ is divisible by 9 for all $n \geq 1$

Step 1: prove true for $n = 1$

$$10^1 + 3(4^{1+2}) + 5 = 10 + 3(64) + 5 \\ = 207$$

which is divisible by 9.

Step 2 Assume true for $n = k$

ie $10^k + 3(4^{k+2}) + 5 = 9P$ where P is a positive integer

Step 3 Prove true for $n = k+1$

ie $10^{k+1} + 3(4^{k+3}) + 5 = 9Q$ where Q is a positive integer.

LHS: $10 \cdot 10^k + 3(4 \cdot 4^{k+2}) + 5$

$$= 10[10^k + 3(4^{k+2}) + 5] - 18(4^{k+2}) - 45$$

$$= 10[9P] - 18(4^{k+2}) - 45$$

$$= 9[10P - 2(4^{k+2}) - 5]$$

$$= 9Q \text{ where } Q = 10P - 2(4^{k+2}) - 5$$

must be a positive integer as P and k are integers).

Hence true for $n = k+1$

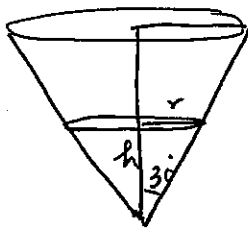
Step 4: Therefore by the principle of mathematical induction, it is true for all $n \geq 1$

Extension 1 Mathematics - Task 3

7th June 2011

SOLUTIONS

Q. 3.
~~(A)~~



$$\frac{dV}{dt} = 48 \text{ cm}^3/\text{min.}$$

$$(i) \quad \frac{r}{h} = \tan 30^\circ$$

$$r = h \tan 30^\circ$$

$$r = \frac{h}{\sqrt{3}}$$

$$\begin{aligned} (ii) \quad V &= \frac{1}{3} \pi r^2 h \\ &= \frac{1}{3} \pi \frac{h^2}{3} \cdot h \\ &= \frac{\pi h^3}{9} \end{aligned}$$

$$\begin{aligned} \frac{dV}{dh} &= \frac{3\pi h^2}{9} \\ &= \frac{\pi h^2}{3} \end{aligned}$$

$$\begin{aligned} \frac{dh}{dt} &= \frac{dV}{dt} \times \frac{dh}{dV} \\ &= 48 \times \frac{3}{\pi h^2} \end{aligned}$$

$$\text{when } h = 12 \text{ cm}$$

$$\frac{dh}{dt} = 48 \times \frac{3}{\pi (12)^2}$$

$$= \underline{\underline{\frac{1}{\pi} \text{ cm/min.}}}$$

Question 4:

$$\begin{aligned}(i) \quad \frac{d}{dt}(We^{kt}) &= e^{kt} \frac{dW}{dt} + W \frac{d}{dt} e^{kt} \\&= e^{kt} [-k(W+10)] + W \cdot k e^{kt} \\&= k e^{kt} [-W - 10 + W] \\&= \underline{-10k e^{kt}} \quad \text{as required}\end{aligned}$$

$$\begin{aligned}(ii) \quad \therefore We^{kt} &= -\int 10k e^{kt} \\&= -10 e^{kt} + C\end{aligned}$$

When $t=0$, $W=15$

$$15 = -10 + C \quad \therefore C = 25$$

$$\therefore We^{kt} = -10 e^{kt} + 25 \quad \div e^{kt}$$

$$W = -10 + 25 e^{-kt}$$

$$\text{or } \underline{W = 25 e^{-kt} - 10}$$

(iii) When $t = 5$ min, $W = 5^\circ\text{C}$

$$(a) \quad 5 = 25 e^{-5k} - 10$$

$$e^{-5k} = \frac{15}{25} = \frac{3}{5}$$

$$k = \frac{\ln \frac{3}{5}}{-5} = 0.1 \quad (1 \text{ dp})$$

(b) When $W = -5^\circ\text{C}$

$$-5 = 25 e^{-kt} - 10$$

$$5 = 25 e^{-kt}$$

$$e^{-kt} = \frac{1}{5}$$

$$t = \frac{\ln \frac{1}{5}}{-k} = 15.75 \dots \approx \underline{\underline{16 \text{ min}}}$$

Q.5

$$P = \$5000$$

$$r = 9\% \text{ pa} : 0.75\% \text{ p. month} \\ = 0.0075$$

$$M = \$500$$

$$(i) A_1 = 5000 (1.0075)^{12} - 500 \\ = \$4969.03.$$

$$(ii) A_2 = [5000 (1.0075)^{12} - 500] (1.0075)^{12} - 500 \\ = 5000 (1.0075)^{24} - 500 (1.0075)^{12} - 500$$

$$A_3 = 5000 (1.0075)^{36} - 500 (1.0075)^{24} - 500 (1.0075)^{12} - 500$$

$\vdots$

$$A_n = 5000 (1.0075)^{12n} - 500 (1.0075)^{12(n-1)} - \dots - 500 (1.0075)^{12} - 500$$

$$= 5000 (1.0075)^{12n} - 500 [1 + 1.0075^{12} + 1.0075^{24} + \dots + 1.0075^{12(n-1)}]$$

$$= 5000 (1.0075)^{12n} - 500 \left[\frac{1 (1.0075^{12n} - 1)}{1.0075^{12} - 1} \right]$$

$$= 5000 \frac{R^{12n} - 1}{R - 1} - 500 \frac{[R^{12n} - 1]}{R - 1}$$

$$(iii) A_n = 0$$

$$0 = \cancel{5000} (1.0075)^{12n} - \cancel{5000} - \frac{500}{1.0075^{12} - 1} (1.0075)^{12n} + \frac{500}{1.0075^{12} - 1}$$

$$\cancel{5000} (1.0075)^{12n} \left(10 - \frac{1}{1.0075^{12} - 1} - 10 \right) = \frac{1}{(1.0075)^{12} - 1}$$

$$1.0075^{12n} = \frac{\frac{1}{(1.0075)^{12} - 1}}{\frac{1}{(1.0075)^{12} - 1} - 10} = x$$

$$12n = \frac{\ln x}{\ln 1.0075}$$

$$\therefore n = \frac{1}{12} \frac{\ln x}{\ln 1.0075}$$

$$= 31.02 \dots \\ \approx 31 \text{ years.}$$